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VIP *Water Harvesting* *Very Important Paper*

Polyzwitterionic Hydrogels for Efficient Atmospheric Water Harvesting

*Chuxin Lei, Youhong Guo, Weixin Guan, Hengyi Lu, Wen Shi, and Guihua Yu**

Abstract: Atmospheric water harvesting (AWH) is regarded as one of the promising strategies for freshwater production desirable to provide sustainable water for landlocked and arid regions. Hygroscopic materials have attracted widespread attention because of their water harvesting performance. However, the introduction of many inorganic salts often leads to aggregation and leakage issues in practical use. Here, polyzwitterionic hydrogels are developed as an effective AWH material platform. Via anti-polyelectrolyte effects, the hygroscopic salt coordinated with polymer chains could capture moisture and enhance the swelling property, leading to a strong moisture sorption capacity. The hydrogel shows superior AWH performance ($0.62\,g\,g^{-1}$, 120 minutes for equilibrium at 30% relative humidity) and produces $5.87\,L\,kg^{-1}$ freshwater per day. It is anticipated that the polyzwitterionic hydrogels with unique salt-responsive properties could provide new insights into the design and synthesis of next-generation AWH materials.

With the increasing population and environmental pollution, freshwater scarcity has become one of the most severe global challenges.[1] Facing these challenges, technologies urgently need to be developed that can obtain clean water from various water resources.[2] Atmospheric water, which is regarded as one of the reliable resources and possesses around 13 sextillions liters, provides the opportunity to harvest water from the ambient air, especially for the remote, landlocked areas and arid regions.[3] Atmospheric water harvesting (AWH) technologies are emerging as sustainable and energy-efficient solutions to provide freshwater.[4] To enable ample water vapor sorption from ambient environments, great efforts have been made to develop state-of-art moisture-capture materials such as hybrid inorganic salts,[5] metal–organic frameworks (MOFs),[6] porous aerogels with hygroscopic salt,[7] natural wood,[8] and hydrogel composites.[4b,9]

Recently, with regards to requirements of all-weather moisture harvesting, employing the hygroscopic salt into the hydrogel matrix is considered a promising strategy to improve moisture-capture performance.[10] Hygroscopic salts firstly capture water vapor and in situ realize liquefaction; then the liquid water is migrated to the hydrogel.[11] The swelling properties of hydrogel facilitate enhancing the water storage, which is beneficial to efficient AWH. However, the high content of salt embedded in the hydrogel matrix may deteriorate the water solubility and swelling property of the hydrogel,[12] which further affects the comprehensive AWH performance. In addition, due to the lack of interactions between salts and macromolecular chains, the salts embedded within hydrogels could easily leak to ambient environments, leading to the deterioration of performance and corrosion of related devices.[13] Therefore, how to effectively incorporate hygroscopic salt without compromising swelling property and stably locked salts within the hydrogel remains challenging.

Zwitterionic polymers represent a novel class of materials and possess oppositely charged cationic and anionic functional groups along their polymer chains. They exhibit unique salt-responsive anti-polyelectrolyte effects, in which salt species can screen self-associate effects between oppositely charged functional groups. The polymer chains tend to show more stretched conformations in the salt solution. Poly(sulfobetaine) possesses better swell capability in salt solutions compared to poly(carboxybetaine) and other polyampholyte materials. The unique salt-responsive property of poly(sulfobetaine) is helpful to coordinate more salt species and achieve more stretched conformations for moisture capture.[14] Here, we report polyzwitterionic hydrogels controllably loaded with hygroscopic LiCl salt for efficient AWH. With the help of the anti-polyelectrolyte effects, hygroscopic salts within the polyzwitterionic hydrogel can not only help capture water vapor but also improve the solubility and swelling properties,[15] leading to the enhancement of water vapor sorption capacity. By rationally tuning the content of LiCl, the as-prepared hydrogel shows a superior water vapor sorption capacity of $0.62\,g\,g^{-1}$ at 30% relative humidity (RH) within 120 minutes. In addition, due to the existence of electrostatic interaction between macromolecular chains and salt, the as-prepared material exhibits impressive stability to enable its long-term practical application. As a proof of concept, we demonstrated the daily production of $5.87\,L\,kg^{-1}$ clean water at 30% RH in a custom-made system. Our results provide new insight into the design of AWH materials by tuning the interactions between hygroscopic salt and polymer chains.

Figure 1a shows the schematic illustration of the synthetic procedures of pure poly [2-

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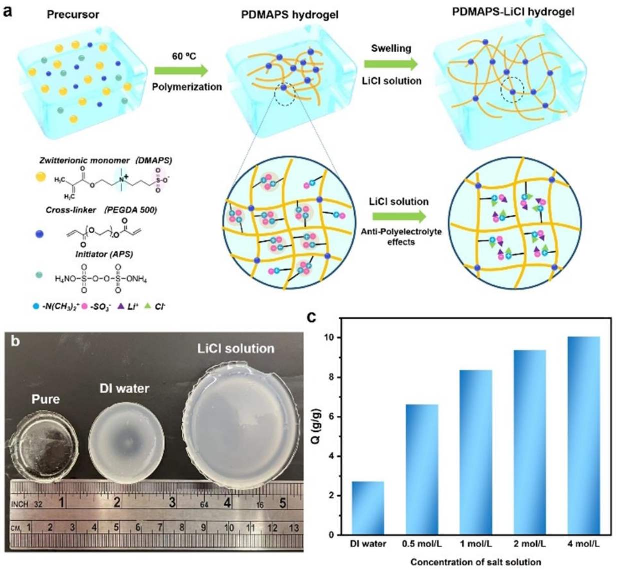

Figure 1. Fabrication and swelling properties of PDMAPS and PDMAPS-LiCl hydrogels. a) Schematic of the fabrication process of PDMAPS and PDMAPS-LiCl hydrogel. b) Optical photograph exhibits the different swelling properties of PDMAPS hydrogel in water and LiCl solution. c) The values of water absorbency of PDMAPS hydrogel swelling in DI water and LiCl solution with different concentrations.

(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide (PDMAPS) hydrogel and hydrogels loaded with LiCl. To begin with, after adding ammonium persulfate (APS) and poly (ethylene glycol)diacrylate (PEGDA) acting as initiator and crosslinker respectively into the DMAPS solution, the hydrogel could be synthesized via radical polymerization under 60 °C around 12 hours (see Supporting Information). The polymer chains tend to exhibit coiled state due to the inter-and intra-attraction of opposite charges. The mobility of the polymer chain is strongly restricted. Based on such anti-polyelectrolyte effects, the polymer chains are likely to show stretched state with more free volumes because the salt species could screen self-association of opposite charges. Compared with the swelling property of hydrogel in deionized (DI) water, it is obvious that hydrogel exhibited a larger swelling ratio in salt solution (Figure 1b). Meanwhile, the water absorbency of hydrogel (Q, [$g\,g^{-1}$]) swelling in a salt solution with different concentrations is represented by the following Equation (1):[16]

$$Q = \frac{m_t - m_0}{m_0} \quad (1)$$

where m_t and m_0 are the weights of the water in the dry state and swollen state at a certain time, respectively. The Q of PDMAPS-LiCl hydrogel swelling in DI water is only 2.7 $g\,g^{-1}$. However, the Q of PDMAPS-LiCl hydrogel swelling in 4 $mol\,L^{-1}$ LiCl solution increases by 9.6 $g\,g^{-1}$ (Figure 1c). Furthermore, the salt sensitivity factor (f) could also be calculated as the following Equation (2):[17]

$$f = 1 - \frac{Q_s}{Q_d} \quad (2)$$

where Q_s and Q_d are the swelling performance in the salt solution and DI water. Conventional hydrogels possess a positive salt sensitivity factor ($0<f<1$),[17] indicating a decreased water solubility in the salt solution. In contrast, the zwitterionic polymer hydrogel behaves negative factors ($f<0$) in different salt solutions (Supporting Information, Figure S1), which points out enhanced swelling properties in the salt solution.

Herein, the AWH performance of PDMAPS-LiCl hydrogel is greatly enhanced due to anti-polyelectrolyte effects. First, the addition of hygroscopic LiCl in the hydrogel matrix could be beneficial to capture more moisture. Second, LiCl also endows the polymer chains with stretched conformation and enhanced mobility, which is conducive to transferring the absorbed water into interiors of hydrogels under low RH environments (Figure 2a). 4 $mol\,L^{-1}$ LiCl solution is used for loading salt into the hydrogel and shows strong moisture capture capacity (Supporting Information, Figure S2). The moisture absorption of hybrid hydrogels was investigated based on a dynamic vapor sorption system (DVS). To achieve efficient freshwater production in one day, a balance between water absorption capacity and kinetics should be emphasized. The moisture sorption capacity of the PDMAPS-LiCl-1 hydrogel is about 0.62 $g\,g^{-1}$ at RH = 30 %. The kinetics is also impressive, which only takes around 120 minutes and 64 minutes to reach equilibrium and 80 % equilibrium, respectively (Figure 2b). The as-prepared hydrogels also exhibit good water sorption capacity under different RH (Supporting Information, Figure S4). Furthermore, the derivative mass change (Supporting Information, Figure S5) also demonstrates that the PDMAPS-LiCl-1 hydrogel exhibited the fastest speed of moisture capture than other samples. The as-prepared hydrogel should be optimal for efficient moisture harvesting for the potential daily water yield.

The sorption–desorption behavior of PDMAPS-LiCl-1 hydrogel is also illustrated (Figure 2c). Although hygroscopic salts in the hydrogel may influence evaporation to some extent, more than 80 % absorbed water (about 0.50 $g\,g^{-1}$) from PDMAPS-LiCl-1 hydrogel could be rapidly released at 80 °C within only 30 minutes, which ensure its efficient water releasing for utilization. With regards to efficient water production for potential practical application, the cycle performance of the PDMAPS-LiCl-1 hydrogel should be concerned. Even after 10 sorption–desorption cycles in the DVS system, the water absorption capacity and kinetics of the PDMAPS-LiCl-1 hydrogel still maintain stable (Figure 2d). Compared with most of the the-state-of-the-art of polymeric sorbent-based AWH materials, the as-prepared polyzwitterionic hydrogel exhibits superior AWH performance.[4b,d,9a,c,10a,11a,c,18]

It is important to understand the relationship between improved AWH performance and salt states and content within the hydrogel. As for the LiCl loaded within the hydrogels, some salt species interact with cationic and

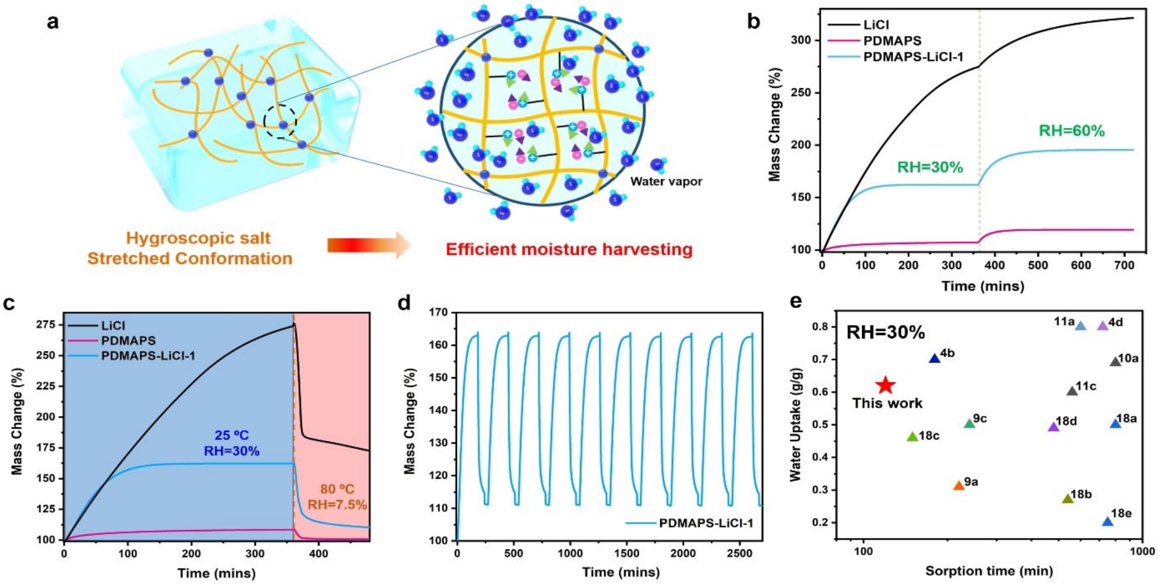

Figure 2. AWH performance of PDMAPS-LiCl-1 hydrogel and other control samples. a) Schematic of the efficient moisture harvesting based on anti-polyelectrolyte effects. The hygroscopic salt and more stretched conformation are both helpful to moisture harvesting. b) The water capturing behavior of LiCl, PDMAPS-LiCl-1 hydrogel, and pure PDMAPS hydrogel at 25 °C, RH = 30 % and RH = 60 %. c) Sorption–desorption curves of LiCl, PDMAPS-LiCl-1 hydrogel, and pure PDMAPS hydrogel at RH = 30 %. d) Sorption–desorption cycle performance of PDMAPS-LiCl-1 hydrogel at RH = 30 %. e) Comparison of water uptake and time required to reach saturated with the state-of-the-art of polymeric sorbent-materials at 30 % RH.

anionic functional groups to screen self-associated effects. The polymer chains are changed to stretched states, which is in good agreement with the swelling property. This part of salt could be called coordinate salt and the rest of the salt without any interaction with polymer chains can be considered as free salt.[19] Due to the limitation of electrostatic interaction with functional groups, the mobility of free salt is much faster than coordinate salt driven by osmotic pressure.[20] Based on different migration rates, most of the coordinate salt stays within the hydrogel, which is reflected by the mass change of swollen PDMAPS-LiCl hydrogel (Supporting Information, Figure S6). Thermogravimetric analyzer (TGA) results (Figure 3a and Supporting Information, Figure S7) reveal that salt content within PDMAPS-LiCl-1 hydrogel decreases to 20 %. In addition, the X-ray diffraction results (XRD) exhibit that the sample with lower salt content didn't show any characteristic peaks of LiCl after removing extra free salt (Figure 3b and Supporting Information, Figure S8). The further Fourier-transform infrared (FTIR) spectroscopy results (Figure 3c and Supporting Information Figure S9) also confirmed the existence of coordinate salt. The peaks at $\approx 1038\,cm^{-1}$ and $\approx 1170\,cm^{-1}$ represent the symmetric and asymmetric stretching vibrations of S=O.[21] As for the PDMAPS-LiCl-1 hydrogel, due to electrostatic interaction between Li^+ and SO_3^-, the symmetric stretching and asymmetric stretching vibrations of S=O shift to $1046\,cm^{-1}$ and $1175\,cm^{-1}$ respectively, indicating that strong interaction exists within the hydrogel. The existence of coordinate salt could better guarantee the stretched conformations of macromolecular chains, which is helpful to promote fast water delivery and water storage within the hydrogel.

The Schematic (Figure 3f) helps to explain the improved AWH performance by controlling the salt content and salt sate within the hydrogel. On the one hand, the evenly distributed, reserved coordinate salt within the hydrogel could rapidly capture moisture as well as maintain stretched conformation for optimal water delivery; on the other hand, in virtue of short-time tuning, most of the aggregated salt could be removed from the hydrogel, which may largely construct the porous structure for highly-efficient water capture and harvesting.

In light of the utilization of PDMAPS-LiCl-1 hydrogel in a practical scenario, we built an atmospheric water releasing and collection system (Figure 4a), which is composed of the fan, water collector, condenser, and flexible heater (Supporting Information, S1.4). The evaporation temperature and condensing temperature could be precisely controlled by the direct current power supply. Meanwhile, the evaporated water could be efficiently condensed in the upper condenser and collected.[4a,22] In conformity with the desorption condition in the DVS system, we still chose 80 °C as desorption temperature. The PDMAPS-LiCl-1 hydrogel could release $0.49\,g\,g^{-1}$ water and collect $0.47\,g\,g^{-1}$ water within 30 minutes at RH = 30 % (Figure 4b). In addition, we also tested cycle performance and actual water production yield per day of PDMAPS-LiCl-1 hydrogel in the water collection system (Figure 4c). To achieve the optimal AWH, we chose to absorb 80 % moisture (60 minutes) and release water (30 minutes) so that more sorption–desorption cycles could be carried out per 24 h period. Thus, $5.87\,L\,kg^{-1}$

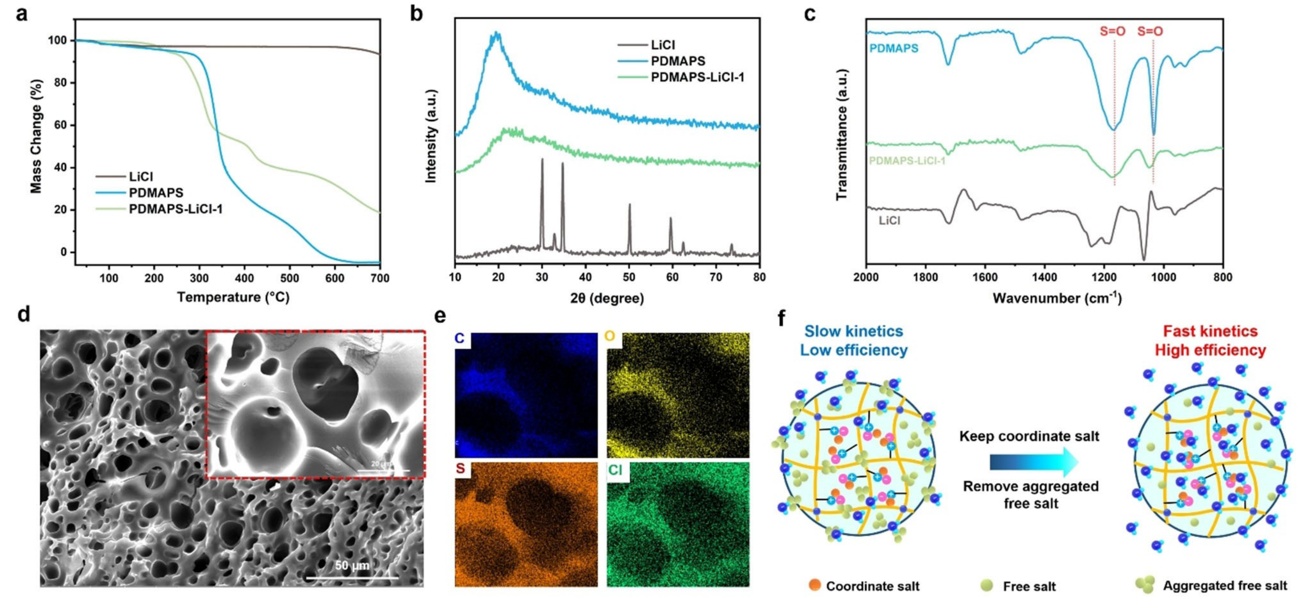

Figure 3. Salt content and states within PDMAPS-LiCl-1 hydrogel corresponding to AWH performance. a) TGA curves of LiCl, dehydrated PDMAPS-LiCl-1 hydrogel, and pure PDMAPS. b) XRD of LiCl, PDMAPS-LiCl-1 hydrogel, and pure PDMAPS. c) FTIR curves of LiCl, PDMAPS-LiCl-1 hydrogel, and pure PDMAPS. d) SEM images of PDMAPS-LiCl-1 hydrogel with different magnifications. e) EDS mapping of PDMAPS-LiCl-1 hydrogel. f) Schematic of removing aggregated free salt as well as keeping coordinated salt for efficient AWH performance.

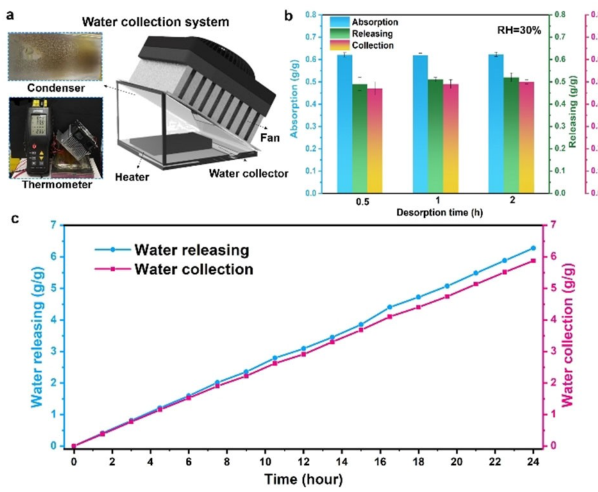

Figure 4. Atmospheric water releasing and collection performance. a) Schematic of water releasing and collection systems. Photographs of the working conditions of the system and condensed clean water droplets on the condenser. b) Water releasing and collection under different desorption times (30 minutes, 1 hour, and 2 hours at 80 °C). c) Freshwater releasing and collection in one day via 16 sorption–desorption cycles.

freshwater could be achieved in one day under RH = 30 % environment. Due to coordinate salt which has strong interaction with functional groups, even after these multiple sorption–desorption cycles in the collection system, the releasing and collection performance of the PDMAPS-LiCl-1 hydrogel still maintain good stability, which indicates its potential long-term utilization for water harvesting and production in practical application.

In summary, we developed a class of polyzwitterionic hydrogels controllably loaded with hygroscopic LiCl for efficient AWH. Owing to anti-polyelectrolyte effects, the added salts could capture water vapor and simultaneously improve the solubility and swelling properties of the hydrogels. The as-prepared hydrogel exhibits an attractive water vapor sorption capacity of 0.62 $g g^{-1}$ at 25 °C, RH = 30 % within 120 minutes. Meanwhile, the as-prepared hydrogel could also produce 5.87 $L kg^{-1}$ freshwater in the practical collection system. Furthermore, owing to the existence of coordinate salts, the PDMAPS-LiCl hydrogel possesses good stability to enable its long-term practical application. This class of polyzwitterionic hydrogels with unique salt-responsive properties can provide new insights to design and prepare next-generation AWH materials for addressing the growing water scarcity challenges.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: Anti-Polyelectrolyte Effects · Atmospheric Water Harvesting · Freshwater Production · Polyzwitterionic Hydrogel

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